



Department of Computer Science and Engineering

Academic Year 2023– 2024 (Odd Semester)

Degree, Semester & Branch: VII Semester – B.E. Computer Science and Engineering

Course Code & Title: CS8792 Cryptography and Network Security

Name of the Faculty member (s): Dr.R.Venkatesh, ASP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / SHA
- **Course Outcome:** CO6
- **Topic Learning Outcome:** TL19
- **Activity Chosen:** One Minute Paper
- **Justification:**
 - In one-minute paper activity the students will be given with 5 minutes to write the brief about that class session.
 - The SHA produces a hash value of fixed length in bits and it is the very basic and important topic needed for Security practices.
 - In order to have a recap of SHA512 logic and Steps, Round Function, and key generation, this activity is chosen.
- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
 - Faculty explained the concept of SHA 1 and SHA-2
 - Students were asked to SHA512 logic and Steps, Round Function, and key generation in a paper within a minute as shown in fig 1.
 - The faculty member collected and corrected the syntax as shown in fig 2. The students who have written wrong procedure were asked to rewrite it.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PSO2
CO2	3	2	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)



• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO2
Justification for correlation	Students will be able to remember the concepts well which is very much needed for a security	Students will be able to apply the concepts while approaching the security practices	Students will be able to provide the solution for the security related problem by applying correct message digest.	As the student involved in the activity as an individual, the individual work skill improved	Students written/ documentation skill will be improved as they are writing the SHA algorithm	The problem solving skill earned through this activity helps the students in solving cyber security problems

• Images / Screenshot of the practice:



Fig 1: One-Minute Paper Activity

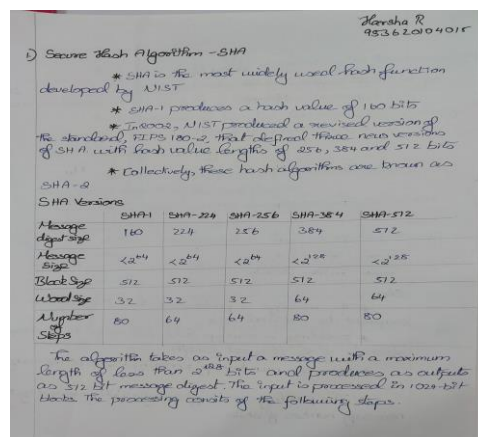


Fig 2: Sample Sheet of activity of the Student Harsha

• Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- Students said that they were able to recollect and remember the concepts of SHA512 logic and Steps, Round Function, and key generation after the activity
- Students felt it will be helpful while using this algorithm

❖ ***Benefit of the practice:***

- Students remembrance of the concept is improved
- Get a sense of what the students have learned, where there might be gaps in their knowledge

❖ ***Challenges faced in implementation:***

- Few students made a mistake in writing the concept.
- It took much time for some students to complete the activity

References:

- <https://www.slideshare.net/raja2dlas/minute-paper>
- <https://www.ritrjpm.ac.in/images/computer-science/CS8251-SE-MinutePaper.pdf>

Signature of Faculty Member

HOD